

Problem Set 7—Suggested Answers

Exercises

Big picture. This problem set develops the *control function approach* to IV estimation. The central idea is elegant: endogeneity arises because e_i is correlated with X_i . We decompose $e_i = v_i'\rho + w_i$ so that w_i is uncorrelated with both X_i and Z_i . Rewriting the structural equation as

$$Y = X\beta + v\rho + w$$

just a projection equation

shows that if we could include v as an additional regressor, OLS would consistently estimate β . The endogeneity has been “absorbed” by v , leaving w as a clean error.

Notation reminder. Throughout, X is $N \times K$, Z is $N \times L$, v is $N \times K$, ρ is $K \times 1$, and w is $N \times 1$. We write $P_A = A(A'A)^{-1}A'$ and $M_A = I - P_A$ for any full-column-rank matrix A .

(a) **Define $\hat{\beta}_0^{\text{OLS}}$ via partitioned regression.**

We regress Y on $[X \ v]$. The regressors are partitioned as $X_1 = X$ (the K variables of interest) and $X_2 = v$ (the K control variables).

Applying the Frisch–Waugh–Lovell (FWL) theorem from the problem sheet with $X_1 = X$ and $X_2 = v$:

$$\hat{\beta}_0^{\text{OLS}} = (X'M_vX)^{-1}X'M_vY,$$

where $M_v = I - v(v'v)^{-1}v' = I - P_v$ is the residual maker that projects out the column space of v .

Interpretation. FWL tells us that $\hat{\beta}_0^{\text{OLS}}$ is numerically identical to the coefficient from regressing $\tilde{Y} := M_vY$ on $\tilde{X} := M_vX$, i.e., from regressing the “ v -residualised” outcome on the “ v -residualised” regressors. In other words, we partial out the linear influence of v from both sides before running OLS.

(b) **Prove that $\hat{\beta}_0^{\text{OLS}} = \beta + \mathbf{o}_p(1)$.**

Step 1: Simplify $\hat{\beta}_0^{\text{OLS}} - \beta$.

Substitute $Y = X\beta + v\rho + w$:

$$\begin{aligned}\hat{\beta}_0^{\text{OLS}} - \beta &= (X'M_vX)^{-1}X'M_vY - \beta \\ &= (X'M_vX)^{-1}X'M_v(X\beta + v\rho + w) - \beta \\ &= (X'M_vX)^{-1}[X'M_vX\beta + X'M_vv\rho + X'M_vw] - \beta \\ &= \mathbf{0}\end{aligned}$$

The first bracket cancels β . The second bracket vanishes because $M_v v = 0$: the matrix M_v projects onto the orthogonal complement of the column space of v , and v itself lies in that column space, so $M_v v = (I - P_v)v = v - v = 0$. Therefore:

$$\hat{\beta}_0^{\text{OLS}} - \beta = (X' M_v X)^{-1} X' M_v w.$$

Step 2: Show $\frac{1}{N} X' M_v w = o_p(1)$.

Expand using $M_v = I - P_v$:

$$\frac{1}{N} X' M_v w = \frac{1}{N} X' w - \frac{1}{N} X' P_v w.$$

First term. By the WLLN, $\frac{1}{N} X' w \xrightarrow{p} E(X_i w_i)$. We claim $E(X_i w_i) = 0$. From the first stage $X_i = \pi' Z_i + v_i$ (row i of $X = Z\pi + v$) and the structural decomposition $e_i = v_i' \rho + w_i$:

- The instrument validity assumption gives $E(e_i Z_i) = 0$, i.e., $\rho' E(v_i Z_i') + E(w_i Z_i') = 0$.
- π is the population OLS projection coefficient in $X = Z\pi + v$, so by construction $E(Z_i v_i') = 0$ (instruments are uncorrelated with the first-stage error).
- Hence $E(w_i Z_i') = 0$.

Therefore:

$$E(X_i w_i) = \pi' E(Z_i w_i) + E(v_i w_i) = 0 + 0 = 0,$$

where the last equality uses $E(v_i w_i) = 0$ (given) and $E(Z_i w_i) = 0$ (just derived).

Second term. Write $X' P_v w = X' v (v' v)^{-1} v' w$ and divide by N :

$$\frac{1}{N} X' P_v w = \underbrace{\left(\frac{1}{N} X' v\right)}_{O_p(1)} \underbrace{\left(\frac{1}{N} v' v\right)^{-1}}_{O_p(1)} \underbrace{\left(\frac{1}{N} v' w\right)}_{o_p(1)},$$

where:

- $\frac{1}{N} X' v \xrightarrow{p} E(X_i v_i') = O_p(1)$ (finite by assumption),
- $\left(\frac{1}{N} v' v\right)^{-1} \xrightarrow{p} (E(v_i v_i'))^{-1} = O_p(1)$ (full rank by assumption),
- $\frac{1}{N} v' w \xrightarrow{p} E(v_i w_i) = 0$, so this factor is $o_p(1)$.

By Slutsky's theorem: $\frac{1}{N} X' P_v w = O_p(1) \cdot O_p(1) \cdot o_p(1) = o_p(1)$.

Combining:

$$\frac{1}{N} X' M_v w = o_p(1) - o_p(1) = o_p(1).$$

Step 3: Denominator.

Under standard rank conditions, $\frac{1}{N} X' M_v X \xrightarrow{p} E(X_i M_v X_i')$, which is positive definite and hence invertible. Its probability limit is $O_p(1)$ with a bounded inverse (also $O_p(1)$).

$$\frac{1}{N} X' M_v X = \frac{1}{N} (X' X - X' P_v X)$$

$$\frac{1}{N} X' X = \frac{1}{N} \sum X_i X_i' = O_p(1)$$

using ~~Q~~ above:

$$\begin{aligned} \frac{1}{N} X' P_v X &= \left(\frac{1}{N} X' v \right) \left(\frac{1}{N} v' v \right)^{-1} \left(\frac{1}{N} X' v \right)' \\ &= O_p(1) \cdot O_p(1) \cdot O_p(1) = O_p(1) \end{aligned}$$

$$\Rightarrow \frac{1}{N} X' M_v X = O_p(1)$$

then by Slutsky's thm

$$\left(\frac{1}{N} X' M_v X \right)^{-1} = O_p(1)$$

Conclusion:

$$\hat{\beta}_0^{\text{OLS}} - \beta = \underbrace{(X' M_v X / N)^{-1}}_{\mathcal{O}_p(1)} \underbrace{(X' M_v w / N)}_{\mathcal{O}_p(1)} = \mathcal{O}_p(1).$$

Hence $\hat{\beta}_0^{\text{OLS}} \xrightarrow{p} \beta$. □

Remark. The proof hinges on $M_v v = 0$ (which kills the ρ term) and $E(v_i w_i) = 0$ (which kills the w term in the second factor of the P_v piece). The assumption $E(v_i w_i) = 0$ is not innocence of v ; it is the *construction* of the decomposition $e = v\rho + w$, which is designed so that w is orthogonal to v .

(c) **Define $\hat{\beta}_1^{\text{OLS}}$ and prove $\hat{\beta}_1^{\text{OLS}} = (X' P_Z X)^{-1} X' P_Z Y$.**

Since v is unobserved we replace it with the first-stage residual

$$\hat{v} = X - Z\hat{\pi} = X - Z(Z'Z)^{-1}Z'X = (I - P_Z)X = M_Z X.$$

We regress Y on $[X \ \hat{v}]$. By FWL:

$$\hat{\beta}_1^{\text{OLS}} = (X' M_{\hat{v}} X)^{-1} X' M_{\hat{v}} Y,$$

where $M_{\hat{v}} = I - P_{\hat{v}}$ and $P_{\hat{v}} = \hat{v}(\hat{v}'\hat{v})^{-1}\hat{v}'$.

The key calculation is to find a simpler expression for $X' M_{\hat{v}}$. Because $\hat{v} = M_Z X$ and M_Z is symmetric and idempotent ($M_Z^2 = M_Z$):

$$\hat{v}'\hat{v} = X' M_Z M_Z X = X' M_Z X.$$

So the projection onto the column space of $\hat{v}$ is:

$$P_{\hat{v}} = M_Z X (X' M_Z X)^{-1} X' M_Z.$$

Pre-multiplying by X' :

$$\begin{aligned} X' P_{\hat{v}} &= X' M_Z X (X' M_Z X)^{-1} X' M_Z \\ &= I_K \cdot X' M_Z \quad (\text{the middle terms cancel}) \\ &= X' M_Z. \end{aligned}$$

Therefore:

$$X' M_{\hat{v}} = X' (I - P_{\hat{v}}) = X' - X' M_Z = X' (I - M_Z) = X' P_Z.$$

Substituting back:

$$\hat{\beta}_1^{\text{OLS}} = (X' M_{\hat{v}} X)^{-1} X' M_{\hat{v}} Y = (X' P_Z X)^{-1} X' P_Z Y. \quad \square$$

This is exactly the 2SLS estimator! The control function approach with estimated first-stage residuals yields the same point estimate as 2SLS. To see the 2SLS structure explicitly, recall $P_Z X = Z(Z'Z)^{-1}Z'X = Z\hat{\pi} =: \hat{X}$ (the fitted values from the first stage). Then:

$$\hat{\beta}_1^{\text{OLS}} = (\hat{X}' X)^{-1} \hat{X}' Y = \hat{\beta}^{\text{2SLS}}.$$

Alternative derivation (expanding the FWL formula directly).

Writing $M_{\hat{v}} = I - \hat{v}(\hat{v}'\hat{v})^{-1}\hat{v}'$ and substituting $\hat{v} = M_Z X$:

$$\begin{aligned}X' M_{\hat{v}} X &= X' X - X' \hat{v} (\hat{v}' \hat{v})^{-1} \hat{v}' X \\&= X' X - X' M_Z X (X' M_Z X)^{-1} X' M_Z X \\&= X' X - X' M_Z X \\&= X' (I - M_Z) X = X' P_Z X, \\X' M_{\hat{v}} Y &= X' Y - X' M_Z X (X' M_Z X)^{-1} X' M_Z Y \\&= X' Y - X' M_Z Y \\&= X' P_Z Y.\end{aligned}$$

The cancellation $X' M_Z X (X' M_Z X)^{-1} = I$ is the core of the argument.

(d) Which estimator is preferred: $\hat{\beta}_0^{\text{OLS}}$ or $\hat{\beta}_1^{\text{OLS}}$?

In terms of consistency, both are equivalent. Part (b) shows $\hat{\beta}_0^{\text{OLS}}$ is consistent. One can show similarly that $\hat{\beta}_1^{\text{OLS}}$ is also consistent (in fact, $\hat{\beta}_1^{\text{OLS}} = \hat{\beta}_1^{\text{2SLS}}$ and consistency of 2SLS follows by the standard argument).

$\hat{\beta}_0^{\text{OLS}}$ is more efficient. The estimator $\hat{\beta}_0^{\text{OLS}}$ uses the *true* first-stage error v , while $\hat{\beta}_1^{\text{OLS}}$ must estimate $\hat{v}$ from the data, introducing additional sampling variation. Replacing v with $\hat{v}$ inflates the asymptotic variance of the estimator for β . As a general principle: using estimated regressors in place of known regressors increases standard errors.

However, $\hat{\beta}_0^{\text{OLS}}$ is infeasible. We never observe the structural error v_i . In practice we are forced to use $\hat{\beta}_1^{\text{OLS}}$. The infeasible $\hat{\beta}_0^{\text{OLS}}$ is a theoretical benchmark.

Bigger picture: the Hausman test. Notice that the coefficient ρ in the augmented regression $Y = X\beta + v\rho + w$ (or its feasible version with $\hat{v}$) directly measures the degree of endogeneity. Testing $H_0 : \rho = 0$ is a test of exogeneity. This is precisely the intuition behind the *Hausman test*: under exogeneity, adding $\hat{v}$ as a control should have a coefficient indistinguishable from zero. If $\hat{\rho}$ is statistically significant, we reject exogeneity of X .

Hausman test of endogeneity

$$y = X\beta + \rho\hat{v} + w$$

regress y on X and $\hat{v}$

test $H_0 : \rho = 0$

rejecting $H_0 \Rightarrow$ endogeneity